

ORIGINAL ARTICLE

PlateChek: a surface-functionalized dielectric microsensor for specific detection of platelet-related hemostatic impairment in whole blood

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Abstract

Background: Platelet count or function defects can result in hemostatic impairment, leading to life-threatening bleeding complications. While platelet transfusions are used to treat such complications, current clinical practice utilizes only platelet count thresholds to guide transfusion. This is because a single precision diagnostic system that can directly correlate both platelet count and function defects to hemostatic deficits is unavailable.

Objectives: We developed PlateChek, a whole blood-based dielectric coagulometry microsensor that uses a gold electrode surface coated with a platelet-specific agonist, and tested the hypothesis that PlateChek can detect hemostatic impairment due to both platelet count and function defects.

Methods: PlateChek incorporates a gold electrode coated with thrombin receptor-activating peptide (TRAP)-6 to render clotting in a platelet-specific manner. Healthy blood was manipulated *in vitro* to recapitulate platelet count or function defects and tested in PlateChek to assess platelet-specific readout signatures. Finally, blood samples from patients with thrombocytopenia or platelet aggregation dysfunctions were used to validate PlateChek's ability to detect clinically relevant hemostatic impairment.

Results: In PlateChek, clotting kinetics were delayed in samples with platelet counts of < 50 k/ μ L compared to healthy controls and clot firmness was reduced with platelet counts of < 100 k/ μ L. PlateChek was sensitive to prolonged clotting time induced by vorapaxar inhibition and reduced clot firmness induced by tirofiban inhibition. PlateChek was sensitive to platelet counts of < 70 k/ μ L and platelet aggregation dysfunctions in clinical blood samples.

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Conclusion: PlateChek is a novel whole blood-based coagulometer with translational potential that directly correlates both platelet count and function defects to hemostatic impairment.

KEYWORDS

biosensing techniques, blood platelets, dielectric spectroscopy, hemostasis, point-of-care testing

1 | INTRODUCTION

Hemostasis is the body's function to stop bleeding via the formation of a blood clot. Platelets perform critical functions in hemostasis, including adhesion to the injury site, activation and aggregation into a platelet plug to stem blood loss, presentation of a procoagulant surface to amplify thrombin generation, and clot retraction via exertion of a tensile force on fibrin (Figure 1A) [1]. Because of the importance of platelets in hemostasis, platelet defects can result in life-threatening bleeding complications. These defects can be quantitative, where platelet count is reduced, or qualitative, where platelet functions are impaired. Many patient groups are at bleeding risk due to platelet defects, with >40% of patients in intensive care settings developing thrombocytopenia (TCP) [2], high prevalence of congenital and acute platelet function defects [3,4], and widespread use of antiplatelet therapies [5]. However, the bleeding phenotype for at-risk groups is nonhomogeneous, with some patients presenting as severe bleeders while others are nonphenotypic [6,7]. This makes diagnostic characterization of hemostatic impairment stemming from platelet defects challenging, since individual markers of platelet count or function can be poor predictors of bleeding phenotype [8,9]. In particular, existing assays such as the rotational thromboelastometry (ROTEM), thromboelastography (TEG), aggregometry, platelet function analysis (PFA), or Total Thrombus-Formation Analysis System (T-TAS) lack the sophistication to directly correlate platelet function defects to bleeding risk. Hence, the de facto consensus is to administer platelet transfusions based on platelet count alone [10], although subjects can have a normal platelet count while actively bleeding [11]. Therefore, diagnostics that broadly and directly assess platelet contributions to overall hemostatic status with sensitivity and specificity can significantly enhance the guidance of platelet transfusions and improve patient outcomes.

To that end, we introduce PlateChek, a platelet-specific microsensor-based coagulometer incorporating gold electrodes that operates on the principles of dielectric spectroscopy (Figure 1B). Dielectric spectroscopy is a measurement that includes a material's capacity to store electrical energy, which changes in blood during clotting [12]. The microsensor incorporates a sputtered-gold electrode surface coated with thrombin receptor-activating peptide (TRAP)-6 (Figure 1C), which activates the platelet protease-activated receptor (PAR)-1, the same receptor activated by thrombin. The microsensor captures the temporal variation in the normalized real part of the complex dielectric permittivity, ϵ'_r , of a coagulating whole

blood sample (<10 μ L) at 1 MHz and provides a coagulation profile with 3 output parameters identified in previous work using uncoated electrodes as follows: T_{peak} , which relates to clotting time [13], $\Delta\epsilon'_{r,max}$, which relates to clot firmness [13], and S_{max} , which relates to fibrin generation rate [14] (Figure 1D). We hypothesized that the novel TRAP-6 coating could render clotting via platelet-specific pathways, making these parameters sensitive to hemostatic impairment caused by both quantitative and qualitative defects in platelets. Therefore, our primary objective was to characterize the sensitivity and specificity of PlateChek to various platelet defects *in vitro* and using clinically relevant samples from patients with known bleeding phenotype.

2 | METHODS

2.1 | Materials

TRAP-6 (sequence SFLLRN), cysteinylated TRAP-6 (sequence SFLLRNC), and cyanine5 (Cy5)-labeled TRAP-6 were ordered as custom peptides with >95% purity (GenScript Biotech). Phosphate-buffered saline (PBS) without calcium and magnesium was purchased from Fisher Scientific (catalog number 10010049). Tirofiban (CAS 150915-40-5) and prostaglandin E1 (CAS 745-65-3) were obtained from Santa Cruz Biotechnology. Calcein AM (catalog number C30990) and phycoerythrin (PE) Annexin V (Catalog number A3511) were purchased from Fisher Scientific. AF647 anti-CD62P was procured from Biolegend (catalog number 304918). Vorapaxar (CAS 618385-01-6) and ninhydrin (CAS 485-47-2) were obtained from Sigma-Aldrich. The ex-tem (catalog number 000503-05-US) and star-tem (catalog number 000503-10-US) reagents as well as cups (catalog number 0002000011) for ROTEM were purchased directly from Werfen. Collagen (catalog number 385) and disposable electrode cuvettes (catalog number 315) for aggregometry were obtained from CHRONO-Log.

2.2 | PlateChek microsensor manufacturing and assembly

The PlateChek microsensor manufacturing process is illustrated in Supplementary Figure S1. Each microsensor consisted of 2 polymethyl methacrylate (PMMA) plastic layers containing gold sensing

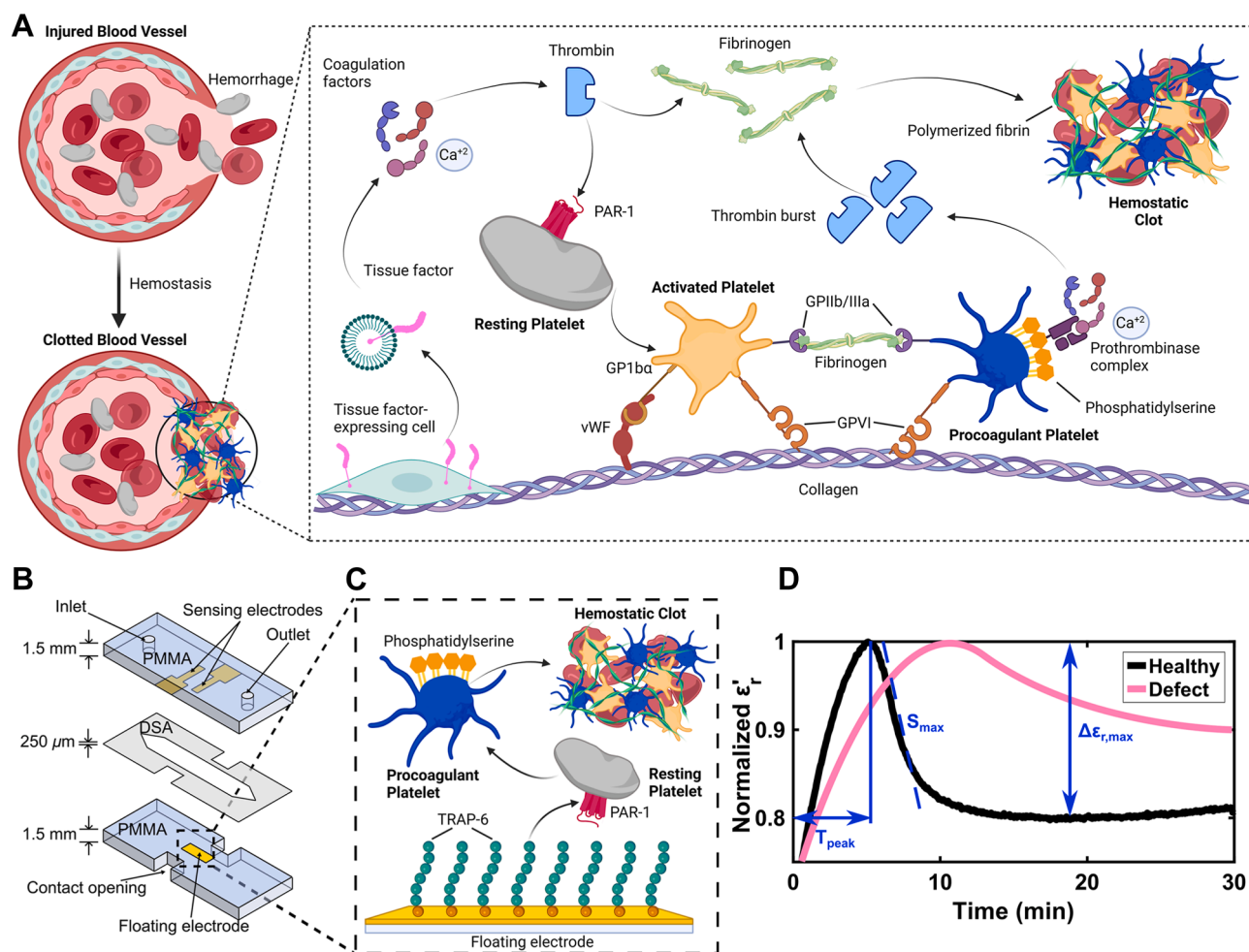


FIGURE 1 (A) Platelets are a critical component of hemostasis. They anchor to an injury site, aggregate to form a plug, and activate to become procoagulant, amplifying generation of thrombin and fibrin. Once fibrin has formed, platelets contract and provide tensile strength to clot structure. (B) Assembly of a PlateChek microsensor. The microsensor consists of 2 polymethyl methacrylate (PMMA) substrates with sputtered-gold sensing and floating electrodes separated by a layer of double-sided adhesive (DSA) film with a cutout for a microfluidic channel. (C) To elicit platelet-specific signal, the floating electrode is coated with cysteinylated thrombin receptor-activating peptide (TRAP)-6, which activates platelet protease-activated receptor (PAR)-1. (D) Characteristic PlateChek coagulation profiles generated at 1 MHz. Three output parameters are extracted from the profile: T_{peak} , which characterizes time to clot formation, $\Delta\epsilon_{r,max}$, which characterizes clot firmness, and S_{max} , which characterizes fibrin generation rate.

and floating electrodes, and an intervening layer of double-sided adhesive (DSA) film with a laser-micromachined cutout to form a microfluidic channel (Figure 1B). Microsensors were fabricated as previously described [15]. Perforated templates for the PMMA layers were machined using a laser cutter (Versa Laser 2.3; Universal Laser Systems Inc.) in arrays of 24 electrodes from ~1.5 mm-thick PMMA sheets. These arrays were washed in 70% ethanol for 1 min, dried manually using non-lint-producing tissue paper, sonicated in deionized (DI) water for 20 min, then dried under ultrapure (99%) nitrogen gas. Next, the arrays were magnet-bound to metal shadow masks and introduced into a sputter system (Denton Discovery 18, Denton Vacuum). The mask cutouts were 4 mm $\times$ 1.5 mm for the floating electrode and 3.5 mm $\times$ 1.3 mm with a spacing of 0.5 mm for the 2 sensing electrodes. Additionally, the sensing electrodes terminated in a rectangular contact area to interface with an Agilent 4294A impedance analyzer (Agilent Technologies, Inc.). Inside the sputtering

chamber, the masked arrays were pre-cleaned for 1 min using 99.99% argon gas under vacuum at 2×10^{-6} torr. The vacuum pressure was maintained throughout the sputtering process. Next, sputtering using a titanium target at 250 W for 62 s with substrate rotation was performed to deposit a ~15 nm-thick titanium adhesion layer. A gold target with 99.99% purity was then used at 250 W for 206 s to deposit a ~200 nm-thick layer for the electrodes. Completed electrodes were stored at room temperature in a desiccation chamber in a class 1000 cleanroom.

A 250 μ m-thick DSA film was laser-micromachined (Versa Laser 2.3) to form the microfluidic channel. The PMMA layers with sputtered-gold sensing and floating electrodes were removed from their perforated templates, aligned, and affixed to each other with the DSA film in between (Figure 1B). For TRAP-6-based functionalization, TRAP-6 was coated onto the floating electrode surface prior to microsensor assembly.

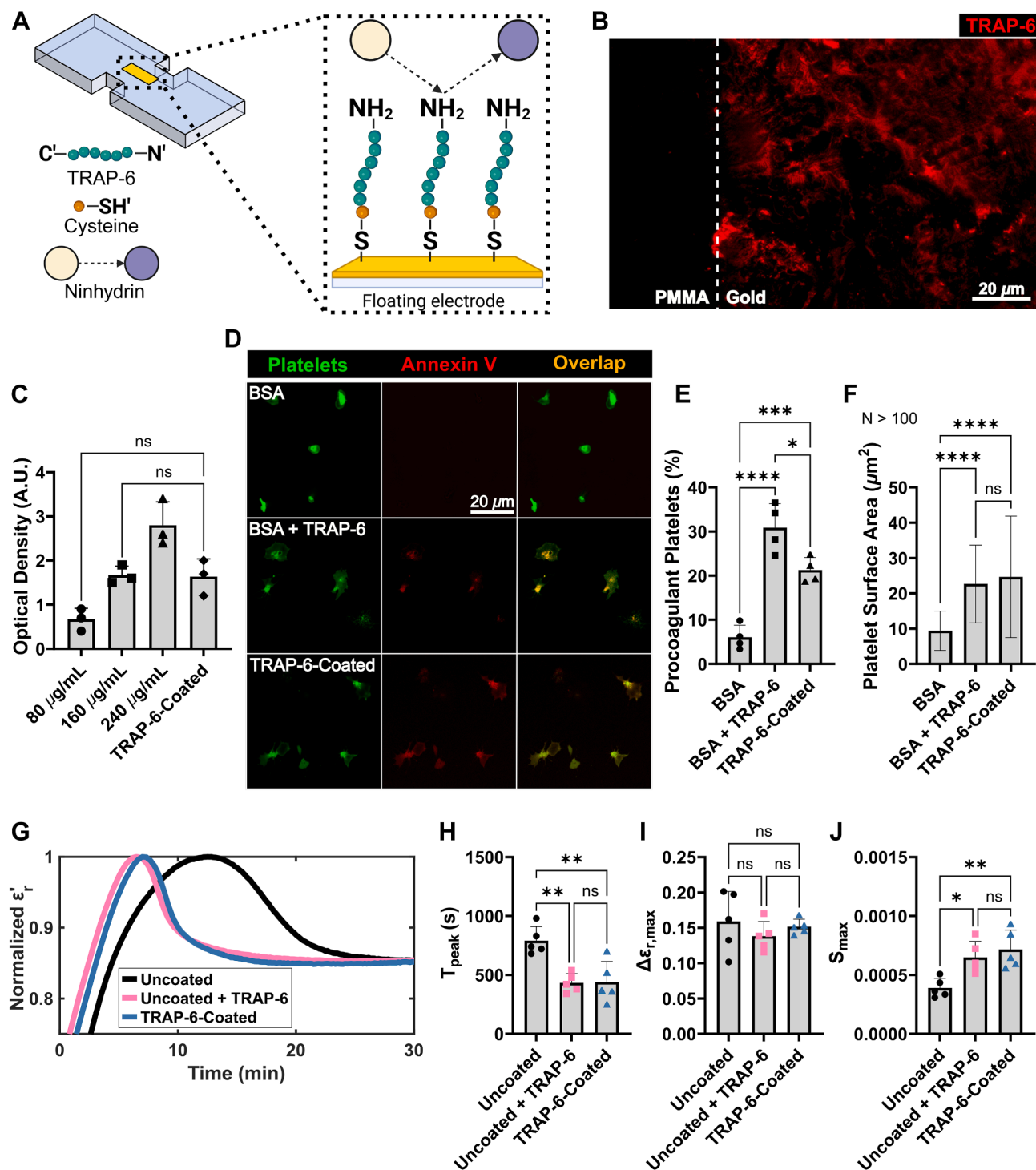


FIGURE 2 (A) Thrombin receptor-activating peptide (TRAP)-6 was modified via C-terminal addition of cysteine to promote interaction between the peptide and the gold floating electrode surface. A colorimetric compound sensitive to N-terminal amine groups was used to determine the amount of cysteinylated TRAP-6 deposited onto the electrode. (B) Fluorescence microscopy image confirming the presence of TRAP-6 on the gold floating electrode surface. (C) Colorimetric optical density for electrodes coated with TRAP-6 compared with that for saline solutions containing soluble-phase TRAP-6. Electrodes coated with TRAP-6 had activity equivalent to ~150- μ g/mL soluble-phase TRAP-6. (D) Confocal microscopy images for platelets on gold electrodes blocked with bovine serum albumin (BSA), in the presence of 150- μ g/mL soluble-phase TRAP-6 or coated with TRAP-6. Platelet spreading and filopodia were evident in the presence of TRAP-6, but not in its absence. Additionally, annexin V staining indicated expression of phosphatidylserine (PS) on platelets activated by soluble-phase TRAP-6 or coated TRAP-6, but not in its absence. (E) TRAP-6-coated electrodes induced expression of PS on 22% $\pm$ 6% of platelets. (F) TRAP-6-coated electrodes induced significant platelet spreading compared with BSA-blocked electrodes. (G) Representative PlateChek coagulation profiles with uncoated electrodes in the absence and presence of 150- μ g/mL soluble-phase TRAP-6 and with TRAP-6-coated electrodes. TRAP-6

2.3 | Preparation of TRAP-6-coated electrodes

Cysteinylated TRAP-6 was commercially obtained from GenScript Biotech to avail the thiol (-SH) group of the cysteine for deposition of TRAP-6 on the gold electrode surface. The cysteine was introduced on the C' terminus, since it has been shown previously that TRAP-6 tolerates C-terminal modification with minimal loss of activity [16,17]. Upon interaction with gold, the sulfhydryl deprotonates, annealing a reversible covalent bond and facilitating immobilization of TRAP-6 [17]. The gold floating electrode was rinsed once with 30% ethanol, then 3 times with DI water. Next, 10 μ L of 2-mg/mL cysteinylated TRAP-6 in a sodium bicarbonate buffer (pH = 10.4) was dispensed onto the electrode surface, and the coated electrode was left to dry overnight at 2 °C in a desiccation chamber. A schematic of the immobilization of TRAP-6 peptide on the gold electrode surface is shown in Figure 2A. TRAP-6-coated electrodes were stored in a refrigerator at 2 to 4 °C where they retained full activity for at least 1 week (Supplementary Figure S2).

2.4 | Colorimetric characterization of TRAP-6 coating on electrodes

To quantify TRAP-6 deposition on coated electrodes, we adapted an assay using the compound ninhydrin, which reacts with N-terminal amine groups on peptides and undergoes a colorimetric complexation reaction [18]. The schematic of this process is shown in Figure 2A. Ninhydrin was dissolved at 2 wt% in ethanol. To generate a standard curve, solutions with soluble-phase TRAP-6 concentrations ranging from 10 μ g/mL to 300 μ g/mL were created. Ninhydrin solution (5% v/v) was added to each sample, and the sample was kept at 80 °C for 10 min. The absorbance at 570 nm was measured using a SpectraMax ID5 plate reader (Molecular Devices) and plotted against TRAP-6 concentration. For coated electrodes, 20 μ L of a solution containing 5% v/v ninhydrin in water was injected into a microsensor with a TRAP-6-coated electrode. The microsensor was held at 80 °C for 10 min. Next, 5 μ L of the injected volume was recovered using a micropipette and dispensed onto a Microvolume Take3 plate (Biotek, working volume 2-10 μ L). The absorbance was measured at 570 nm, and the TRAP-6 deposition was back-calculated using the standard curve.

2.5 | Fluorescence imaging-based confirmation of TRAP-6 coating on electrodes

Fluorescently-tagged TRAP-6 labeled with Cyanine5 was used to verify successful immobilization of TRAP-6 on the gold electrode.

Fluorescently-tagged TRAP-6 was incubated on the floating electrode overnight in a sodium bicarbonate buffer (pH = 10.4). Next, the electrode was washed 3 times with the sodium bicarbonate buffer and imaged under epifluorescence using a Zeiss Axio Observer (Carl Zeiss) using a 40 $\times$ objective (Plan Neo-FLUAR, NA 1.3) with an excitation bandpass filter at 594 nm and emission bandpass filter at 647 nm.

2.6 | Blood sample procurement

Blood samples from de-identified, healthy donors were obtained under an Institutional Review Board (IRB)-approved protocol (IRB# 09-90-195) from the Hematopoietic Biorepository and Cellular Therapy Core at Case Western Reserve University. Healthy donors were defined as those who had not been diagnosed with any bleeding disorder, had not taken anticoagulant or antiplatelet medications, did not have hepatitis, and had not used nonsteroidal anti-inflammatory drugs (NSAIDs) for at least 2 weeks prior to blood donation. Additionally, only blood samples from donors with platelet counts of > 150 k/ μ L were used for this study. Pediatric patient samples were collected under a separate IRB-approved protocol at the University Hospitals Rainbow Babies and Children's Hospital (IRB# STUDY20210247), where they were receiving treatment for previously diagnosed bleeding abnormalities. Blood counts and aggregometry assays were performed to verify presence of a blood pathology related to either platelet count or function defects. Patients with additional coagulation abnormalities other than platelet count or function defects were excluded. All blood samples were drawn by venipuncture into 3.2% sodium-citrate anticoagulant vacutainers. Samples were collected immediately after donation and carried to our research laboratory, which was within a 5-min walk from both sample-collection sites. All assays were completed within 3 hours of blood collection.

2.7 | Fluorescence imaging-based characterization of procoagulant platelet formation on TRAP-6-coated electrodes

Citrated whole blood was centrifuged at 150g for 15 min at 25 °C to isolate platelet-rich plasma (PRP). Next, 10%v/v acid citrate dextrose (ACD) solution (38-mM citric acid, 75-mM trisodium citrate, and 136-mM dextrose in Tyrode's buffer) was added to the PRP fraction, and prostaglandin E₁ was added to a final concentration of 1 μ M. Platelets were then centrifuged at 2500g for 25 min at 25 °C. The plasma supernatant was discarded, and the washed

reduced T_{peak} and increased S_{max} without affecting $\Delta\epsilon_{r,\text{max}}$ in healthy samples. There was no difference between curves generated with soluble-phase TRAP-6 or coated TRAP-6. (H) T_{peak} was significantly reduced with TRAP-6-coated electrodes. (I) $\Delta\epsilon_{r,\text{max}}$ was not affected by TRAP-6-coated electrodes. (J) S_{max} was significantly increased with TRAP-6-coated electrodes. Statistics are shown as mean $\pm$ SD; ns, not significant; * P < .05; ** P < .01; *** P < .001; **** P < .0001.

platelets were resuspended in PBS. Next, 1 $\mu\text{g/mL}$ of Calcein AM in dimethylsulfoxide (DMSO) was added to stain the platelets. Additionally, either AF647 anti-CD62P (to stain for P-selectin and probe platelet activation) or phycoerythrin annexin V (to stain for phosphatidylserine [PS] and probe procoagulant induction) was added at 2% v/v. Uncoated floating electrodes were blocked with 1-wt% bovine serum albumin (BSA) for 60 minutes. Next, 10 μL of the washed platelets were incubated on BSA-blocked electrodes in the absence or presence of soluble-phase TRAP-6 or on TRAP-6-coated electrodes for 15 minutes at 37 $^{\circ}\text{C}$. To prevent drying, the electrodes were kept in a sealed Petri dish with a moist Kimwipe during incubation. After incubation, 2 μL of 3-wt% paraformaldehyde was added to fix the samples and prevent further platelet activation or procoagulant induction. Samples were washed 3 times using 10 μL of PBS to remove excess fluorophores, then imaged using a Stellaris 5 confocal microscope (Leica). To quantify the percentage of activated or procoagulant platelets, images were captured using a 20 $\times$ objective (Plan APO, NA 0.75). High-resolution and co-localization images were captured using a 63 $\times$ oil immersion objective (Plan APO, NA 1.4).

2.8 | *In vitro* preparation of blood samples to simulate platelet count and function defects

Platelet count-deficient blood samples to simulate TCP were generated using a blood separation and reconstitution protocol (Figure 3A). Citrated whole blood was centrifuged at 150g for 15 minutes at 25 $^{\circ}\text{C}$. PRP supernatant was collected without disturbing the buffy coat. Next, the buffy coat was discarded, and the red blood cell (RBC)-rich pellet was collected. Half of the PRP was further centrifuged at 13 000g for 5 minutes at room temperature to remove the platelets and isolate platelet-free plasma. The RBC pellet was resuspended in an equal volume of isotonic PBS and then centrifuged at 500g for 5 minutes. This wash step was repeated 3 times to remove entrained platelets and plasma from the RBC fraction. After the final wash step, the supernatant was removed but the pellet was not resuspended, resulting in a rich RBC fraction. A DxH 500 Complete Blood Counter (Beckman Coulter) was used to quantify the platelet count and hematocrit in the platelet-free plasma, PRP, and RBC fractions. Based on the measured results, the fractions were recombined into reconstituted whole blood with normal hematocrit

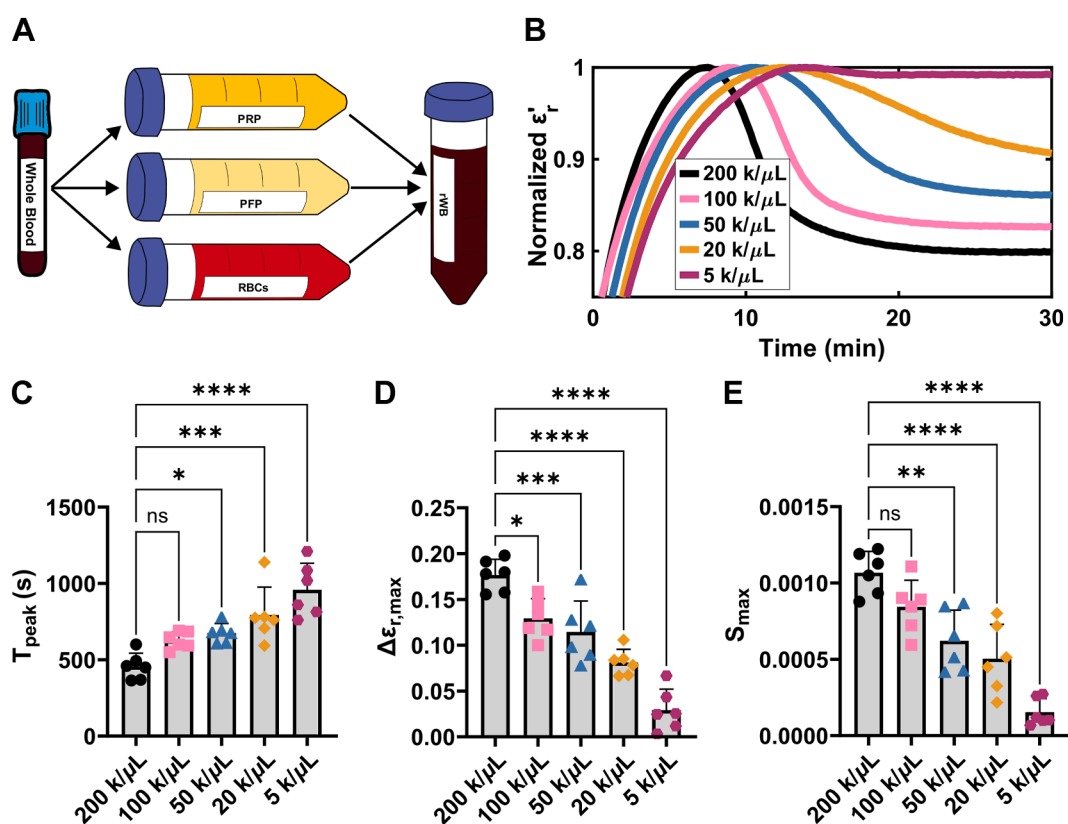


FIGURE 3 Thrombin receptor-activating peptide (TRAP)-6-enhanced dielectric coagulometry was sensitive to platelet count defects. (A) Healthy donor blood was separated into its constituent components and then recombined to recapitulate platelet count defects *in vitro* without affecting coagulation factor concentrations or hematocrit. (B) Representative PlateChek coagulation profiles for reconstituted whole blood (rWB) samples with decreased platelet counts. Reducing platelet count prolonged T_{peak} and reduced $\Delta\epsilon_{r,\text{max}}$. (C) Prolongation in T_{peak} was significant at platelet counts of ≤ 50 k/ μL ($\leq 25\%$ of healthy levels), but not at 100 k/ μL . (D) $\Delta\epsilon_{r,\text{max}}$ decreased at all reduced platelet counts including at 100 k/ μL . (E) Decrease in S_{max} was significant at platelet counts of ≤ 50 k/ μL , but not at 100 k/ μL . Statistics are shown as mean $\pm$ SD; ns, not significant; * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

and plasma levels, but variable platelet counts. Our target hematocrit was 40%, and our target platelet counts ranged from 5 k/ μ L (simulating TCP) to 200 k/ μ L (simulating a healthy physiological platelet count). DxH 500 was used to verify the composition of reconstituted samples. A 10% deviation from target values was tolerated; samples with >10% deviation were readjusted or discarded.

Platelet function defects in patients can also result in pathologic outcomes [19]. To generate blood samples with platelet function defects, citrated whole blood samples were confirmed to have a healthy platelet count (> 150 k/ μ L) and hematocrit (30%-40%) using DxH 500. Samples were then treated with either vorapaxar (final concentration = 50 nM) or tirofiban (final concentration = 100 nM) (Figure 4A). Vorapaxar, which has clinical relevance as an antiplatelet therapy [20,21], results in a platelet activation defect via blocking PAR-1 receptors. In contrast, tirofiban induces a defect in platelet aggregation and clot contraction via competitive inhibition of glycoprotein (GP)IIb/IIIa receptors [22]. This mimics the genetic disorder Glanzmann thrombasthenia and recapitulates GPIIb/IIIa receptor inactivation that can occur because of trauma-induced coagulopathy or antibodies generated due to infection [23–25].

2.9 | PlateChek microsensor measurements

Measurements on the PlateChek microsensor were conducted using an Agilent 4294A impedance analyzer (Agilent Technologies Inc.) within a thermostatic chamber maintained at 37 °C. Citrated whole blood samples were prewarmed to 37 °C and recalcified to 20-mM CaCl_2 . Immediately after recalcification, 20 μ L of the mixture was injected into the PlateChek microsensor to fill the microfluidic channel volume (<10 μ L). Excess blood was absorbed into a Kimwipe. The dielectric permittivity of the coagulating blood sample was then measured over a frequency range of 10 kHz to 100 MHz. The PlateChek readout curve was defined as the temporal variation in the normalized real part of the complex dielectric permittivity, ϵ_r' , of the coagulating blood sample at 1 MHz. All measurements were conducted in technical duplicate.

2.10 | Multidimensional analysis of PlateChek readout parameters

Data from multiple PlateChek readout parameters were considered simultaneously in multidimensional space to generate more precise diagnostic information than what could be obtained from any single parameter. For this approach, we ensured that each parameter (ie, T_{peak} , $\Delta\epsilon_{r,\text{max}}$, and S_{max}) was weighted equally during analysis. Since each parameter occupied a different dynamic range (100–1500 seconds for T_{peak} ; 0.01–0.2 for $\Delta\epsilon_{r,\text{max}}$; and 0.0001–0.0015 for S_{max}), we defined a defect ratio for each parameter to give them equivalent numerical weighting. The defect ratio was calculated for each data point and defined as the measured experimental value for that datum divided by the healthy average for that parameter. Therefore, a

defect ratio of 1 was indicative of a sample with a healthy coagulation profile. Ratios other than 1 (whether higher or lower) indicated abnormal coagulation. The defect ratios for multiple parameters were then plotted against each other in multidimensional space, making it possible to differentiate diagnostic “fingerprints” for distinct platelet defects. For each defect ratio, 95% confidence regions were calculated by adding or subtracting 2 SDs to/from the mean.

3 | RESULTS

3.1 | Design and characterization of TRAP-6-coated electrodes

To render clotting in a platelet-specific manner, we developed electrodes coated with TRAP-6. TRAP-6 was chosen as the platelet agonist because the coagulation profile readout parameters were more sensitive to its effects on healthy blood than those of either adenosine diphosphate or collagen in our preliminary studies (Supplementary Figure S3). The process for coating electrodes with TRAP-6 and its assessment using the ninhydrin assay (Figure 2A) have been described previously in the Methods section. Figure 2B shows the representative fluorescence image of a gold electrode coated with Cy5-labeled cysteinylated TRAP-6, confirming TRAP-6 immobilization on the electrode. Figure 2C shows the ninhydrin-based quantification analysis of electrode-immobilized TRAP-6, indicating that electrodes bore an effective TRAP-6 concentration of $\sim 150 \mu\text{g/mL}$. This concentration falls in the range of soluble-phase TRAP-6 used in aggregometry and PFA assays [26,27] and was sufficient to maximize the effect of TRAP-6 on healthy blood in PlateChek (Supplementary Figure S4). The mechanism of action by which TRAP-6-coated electrodes induced platelet-specific response was studied using confocal fluorescence microscopy. As shown in Supplementary Figure S5, platelet adhesion increased significantly on TRAP-6-coated electrodes compared with BSA-blocked electrodes. P-selectin expression quantified by CD62P staining indicated that the percentage of electrode-adhered platelets expressing P-selectin increased from $7.3\% \pm 3.2\%$ on BSA-blocked electrodes to $38.0\% \pm 9.2\%$ on TRAP-6-coated electrodes (Supplementary Figure S6). Figure 2D shows representative images of PS expression analyzed on TRAP-6-coated electrodes compared with controls. Figure 2E shows quantification of PS expression from such studies, indicating that the percentage of procoagulant platelets increased from $5.5\% \pm 3.8\%$ on BSA-blocked electrodes to $20.2\% \pm 4.1\%$ on TRAP-6-coated electrodes. Additionally, the per-platelet surface area increased from $< 10 \mu\text{m}^2$ on BSA-blocked electrodes to $> 20 \mu\text{m}^2$ on TRAP-6-coated electrodes (Figure 2F), a procoagulant surface area increase that was accompanied by increased filopodia generation as evidenced in Supplementary Figure S7. Altogether, these results indicate that the gold electrodes could be effectively coated with TRAP-6, which in turn resulted in significant platelet adhesion, activation, and generation of procoagulant platelet surface area.

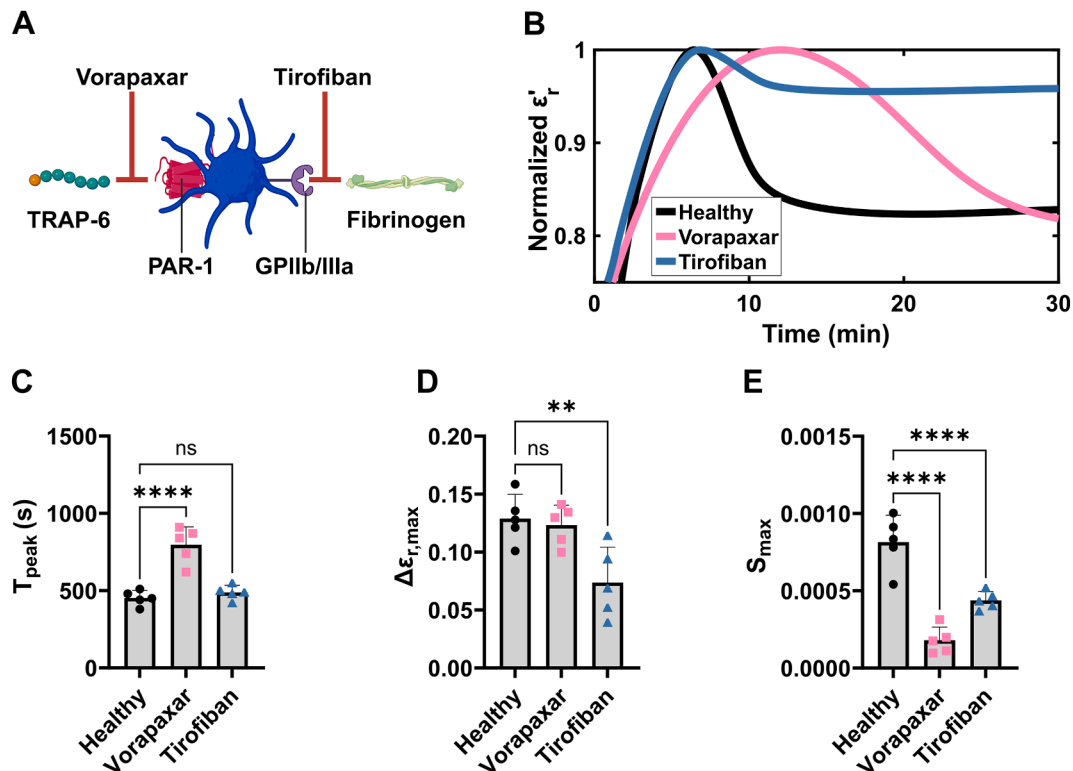


FIGURE 4 Thrombin receptor-activating peptide (TRAP)-6-enhanced dielectric coagulometry was sensitive to platelet function defects. (A) Healthy blood was treated with vorapaxar, a protease-activated receptor (PAR)-1 receptor antagonist, or tirofiban, a glycoprotein (GP) IIb/IIIa receptor antagonist, to inhibit platelet function *in vitro*. (B) Representative PlateChek coagulation profiles for healthy and inhibited samples, showing sensitivity to the effects of both inhibitors. Vorapaxar prolonged T_{peak} , whereas tirofiban reduced $\Delta\epsilon_{r,max}$. (C) Vorapaxar, but not tirofiban, significantly prolonged T_{peak} . (D) Tirofiban, but not vorapaxar, significantly reduced $\Delta\epsilon_{r,max}$. (E) Both vorapaxar and tirofiban significantly reduced S_{max} . Statistics are shown as mean $\pm$ SD; ns, not significant; * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

3.2 | Effect of TRAP-6-coated electrodes on coagulation profile of healthy blood

Figure 2G shows representative coagulation profiles of healthy human whole blood tested on uncoated electrodes, uncoated electrodes with TRAP-6 added to the blood sample, and TRAP-6-coated electrodes. Figure 2H–J show quantitative analysis of the specific coagulometry readout parameters T_{peak} , $\Delta\epsilon_{r,max}$, and S_{max} extracted from such coagulation profiles. As evident from the data, on uncoated electrodes, clotting time in healthy blood was characterized by a T_{peak} of 12 to 15 minutes, whereas 100 $\mu\text{g/mL}$ of soluble-phase TRAP-6 or TRAP-6-coated electrodes accelerated clotting kinetics and reduced T_{peak} to 6 to 7 minutes (Figure 2H). TRAP-6 did not impact the clot firmness parameter $\Delta\epsilon_{r,max}$ (Figure 2I), indicating that TRAP-6 did not influence the platelet-mediated fibrin contraction. However, fibrin generation rate was significantly accelerated in the presence of TRAP-6, as evident from the 2-fold increase of S_{max} on TRAP-6-coated electrodes (Figure 2J). Collectively, these results in conjunction with the platelet analysis results described in the previous section indicate that the significant adhesion, activation, and procoagulant induction of platelets on the TRAP-6-coated electrodes led to amplified thrombin generation, which in turn accelerated clotting kinetics via a platelet-specific pathway. This indicates the

potential of PlateChek with TRAP-6-coated electrodes in correlating platelet status with hemostatically important mechanistic and kinetic aspects of blood clotting.

3.3 | Sensitivity of TRAP-6-coated electrodes to TCP

The most prevalent platelet defect that leads to hemostatic impairment is TCP (ie, reduced platelet count). Therefore, TCP with varying severity was simulated *in vitro* in reconstituted whole blood as described previously (see Methods, schematic in Figure 3A). On TRAP-6-coated electrodes, the measured coagulation profile was significantly affected by the platelet count, as evident from the representative curves shown in Figure 3B. Quantitative analysis revealed that T_{peak} increased with reduced platelet counts (Figure 3C), while $\Delta\epsilon_{r,max}$ (Figure 3D) and S_{max} (Figure 3E) diminished. Interestingly, the prolongation of T_{peak} and the reduction in S_{max} became statistically significant only at platelet counts of $\leq 50 \text{ k}/\mu\text{L}$. This number has specific clinical relevance, since a platelet count of $< 50 \text{ k}/\mu\text{L}$ can be an indication for platelet transfusion in patients undergoing surgery or benefiting from ongoing care after traumatic injuries [28,29].

To understand the diagnostic advantages of using TRAP-6-coated electrodes, we performed identical experiments using uncoated electrodes. With uncoated electrodes, reduced platelet counts had no effect on T_{peak} (Supplementary Figure S8A), caused a decrease in $\Delta\epsilon_{r,\text{max}}$ (Supplementary Figure S8B), and weakly affected S_{max} with only severe TCP generating a statistically significant reduction (Supplementary Figure S8C). Therefore, compared with uncoated electrodes, TRAP-6-coated electrodes offered superior, multi-parametric, platelet count-specific sensitivity to clotting kinetics and clot firmness. We also compared data from PlateChek with TRAP-6-coated electrodes to those from clinical assays, namely, ROTEM and aggregometry, and identified a strong correlation between certain parameters (Supplementary Figure S9), indicating that data from PlateChek could be clinically meaningful.

3.4 | Sensitivity of TRAP-6-coated electrodes to platelet function defects

To test the sensitivity of TRAP-6-coated electrodes to platelet function defects, we used platelet PAR-1 inhibition with vorapaxar and GPIIb/IIIa inhibition with tirofiban, as described previously (see Methods, schematic in Figure 4A). TRAP-6-coated electrodes in PlateChek exhibited sensitivity to both vorapaxar and tirofiban inhibition, as evident from representative profiles shown in Figure 4B. Based on the quantitative analysis shown in Figure 4, vorapaxar prolonged T_{peak} (Figure 4C), did not impact $\Delta\epsilon_{r,\text{max}}$ (Figure 4D), and reduced S_{max} (Figure 4E). These effects are intuitive since vorapaxar inhibited the PAR-1 receptor, effectively abrogating platelet interactions with the electrode-immobilized TRAP-6 and thus extending T_{peak} to 12 to 15 minutes. In fact, this timeframe is equivalent to the range for T_{peak} on uncoated electrodes (Supplementary Figure S10A). It is also important to note that uncoated electrodes were insensitive to vorapaxar-induced platelet inhibition, highlighting another advantage of TRAP-6-coated electrodes (Supplementary Figure S10).

In comparison with vorapaxar, platelet GPIIb/IIIa inhibition via tirofiban did not affect the ability of TRAP-6-coated electrodes to activate platelets, such that T_{peak} was unaffected (Figure 4C). Tirofiban instead prevented robust interactions between platelets themselves (ie, platelet-fibrinogen interaction) and between platelets and fibrin (ie, clot contraction), resulting in a weaker clot as evidenced by reduced $\Delta\epsilon_{r,\text{max}}$ (Figure 4D). Surprisingly, tirofiban also reduced S_{max} (Figure 4E). A possible explanation for this is that platelet spreading is mediated in part by GPIIb/IIIa receptors [30]; tirofiban could therefore inhibit platelet spreading, reducing pro-coagulant surface area and leading to reduced fibrin generation rate. Uncoated electrodes were sensitive to tirofiban-induced platelet inhibition only in $\Delta\epsilon_{r,\text{max}}$, but not in T_{peak} or S_{max} (Supplementary Figure S10). As with TCP samples, PlateChek data generated from platelet-dysfunction samples correlated well with both ROTEM and aggregometry (Supplementary Figure S11).

3.5 | Sensitivity of TRAP-6-coated electrodes to multiple concomitant platelet defects

Clinically, it is not uncommon for patients to present with multiple concomitant platelet defects. For example, congenital diseases, hemorrhagic shock, or trauma-induced coagulopathy often manifest as a combination of TCP with platelet function defects [25,31,32]. Therefore, we sought to evaluate whether TRAP-6-enhanced dielectric coagulometry can detect hemostatic impairment stemming from concomitant platelet defects and, furthermore, determine whether cumulative data across multiple parameters can be used to differentiate between distinct platelet defect types. To that end, we created several concomitant defect types. For some cases, we created TCP samples with a platelet count of 20 k/ μ L and further induced platelet function inhibition in the samples using vorapaxar or tirofiban. In other conditions, samples with healthy platelet counts were treated with a combination of vorapaxar and tirofiban to induce a double defect in platelet function. For the severest condition, TCP samples were treated with a combination of vorapaxar and tirofiban to induce a triple defect.

We found that the effects of individual defects were “additive.” For example, while tirofiban alone caused a decrease in $\Delta\epsilon_{r,\text{max}}$ and S_{max} but no impact on T_{peak} (Figure 4C–E) and vorapaxar alone caused prolonged T_{peak} and reduced S_{max} with no effect on $\Delta\epsilon_{r,\text{max}}$ (Figure 4C–E), the combination of tirofiban with vorapaxar caused prolonged T_{peak} , reduced $\Delta\epsilon_{r,\text{max}}$, and a severely reduced S_{max} (Figure 5A–C). The most severe impairments, which are likely predictive of bleeding risk, were associated with TCP in combination with platelet function defects. For example, the combination of TCP with vorapaxar resulted in the greatest prolongation in T_{peak} of any 2-pronged model (Figure 5A). Similarly, TCP with tirofiban caused the most severe reduction in $\Delta\epsilon_{r,\text{max}}$ (Figure 5B). Finally, TCP with vorapaxar significantly knocked down S_{max} (Figure 5C). Representative coagulation profiles from these studies are shown in Supplementary Figure S12.

3.6 | Diagnostic differentiation between platelet defects using multidimensional analysis

Since each platelet defect generated a unique effect on the PlateChek readout parameters, we hypothesized that it might be possible to differentiate between platelet defect types to enhance diagnostic specificity. Therefore, we developed an analytical method to synthesize the PlateChek readout parameters in multidimensional space using their “defect ratios.” For example, in the 2-dimensional (2D) representation of data where T_{peak} and $\Delta\epsilon_{r,\text{max}}$ define the axes, distinct regions emerged for different platelet defects (Figure 5D). The black oval indicates the 95% confidence region for healthy samples. In comparison, tirofiban-inhibited samples (purple) occupied a space lower on the ordinate axis, and vorapaxar-inhibited samples (blue) occupied a distinct space to the right on the abscissa.

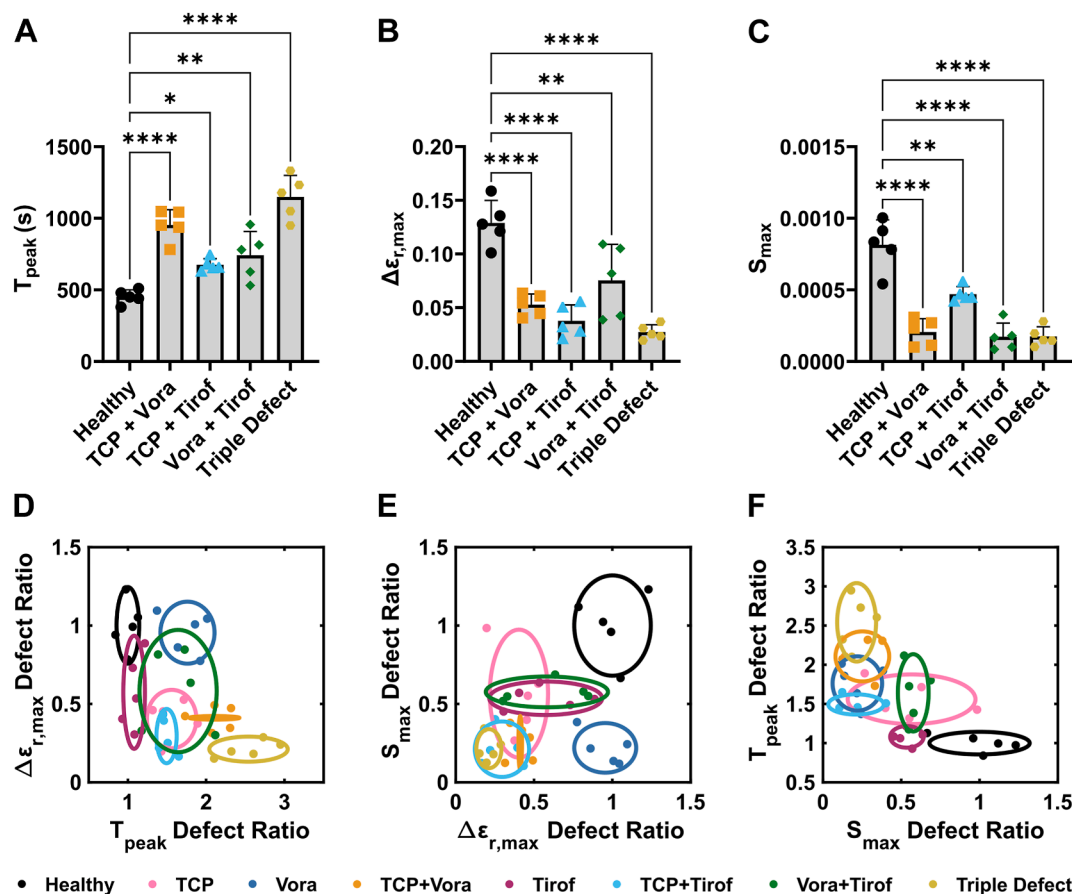


FIGURE 5 Thrombin receptor-activating peptide (TRAP)-6-enhanced dielectric coagulometry was sensitive to multiple concomitant platelet defects. (A) Prolongation in T_{peak} caused by thrombocytopenia (TCP) + Vora was more severe than that caused by each defect alone as seen in Figures 3C and 4C. (B) Decrease in $\Delta\epsilon_{r,max}$ caused by TCP + Tirof was more severe than that caused by each defect alone as seen in Figures 3D and 4D. (C) All combination defects significantly reduced S_{max} . (D–F) Multiparametric analysis comparing the effects of each type of defect. A defect ratio of 1 indicates that the sample behaved equivalent to the healthy control. Values other than 1 (either higher or lower) indicate impairment. Ovals indicate 95% CIs. (D) Distinct regions were evident when plotting $\Delta\epsilon_{r,max}$ against T_{peak} , permitting distinction between different types of impairment. For example, vorapaxar affected only T_{peak} , tirofiban affected only $\Delta\epsilon_{r,max}$, and TCP affected both. Combinations of defects caused more pronounced shifts in defect ratios than single defects alone. The triple defect was characterized by a severe shift in both parameters. (E) Multiparametric analysis using S_{max} and $\Delta\epsilon_{r,max}$ enabled better distinction between certain groups such as TCP and TCP + Tirof. (F) Multiparametric analysis using T_{peak} and S_{max} showed partial separation between most groups. Statistics are shown as mean $\pm$ SD; ns, not significant; * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

TCP samples (pink) occupied another space shifted diagonally from the healthy region (Figure 5D). In this 2D representation, some groups could not be differentiated; eg, dual inhibition by vorapaxar and tirofiban (green) resulted in a region that overlapped significantly with that for TCP samples treated with tirofiban (cyan). However, if the axes were defined using $\Delta\epsilon_{r,max}$ and S_{max} , these groups occupied distinct regions that did not overlap (Figure 5E), highlighting the potential of this multidimensional analytical approach. Defining the axes using S_{max} and T_{peak} , the third and final possible 2D representation, provided yet another distribution with unique data separation (Figure 5F). Elevating this approach to 3D space resulted in a representation with minimal overlap between groups (Supplementary Figure S13), suggesting that it might be possible to generate highly specific diagnostic information using TRAP-6-enhanced dielectric coagulometry.

3.7 | Validation of PlateChek using pediatric patient blood samples

Building on the promising results obtained using *in vitro* models of TCP and platelet function defects with the PlateChek microsensor featuring a TRAP-6-coated electrode, we sought to validate PlateChek using blood samples collected from patients with platelet defects. To that end, de-identified citrated blood was collected from pediatric patients with TCP or congenital platelet function defects. Samples from TCP patients were binned according to the severity of TCP using clinically relevant platelet count thresholds (healthy, count > 150 k/ μ L; mild, 70 k/ μ L $<$ count ≤ 150 k/ μ L; moderate, 20 k/ μ L $<$ count ≤ 70 k/ μ L; severe, count ≤ 20 k/ μ L). In PlateChek, the coagulation profile became increasingly abnormal with increasing severity of TCP (Figure 6A). All 3 PlateChek readout parameters were

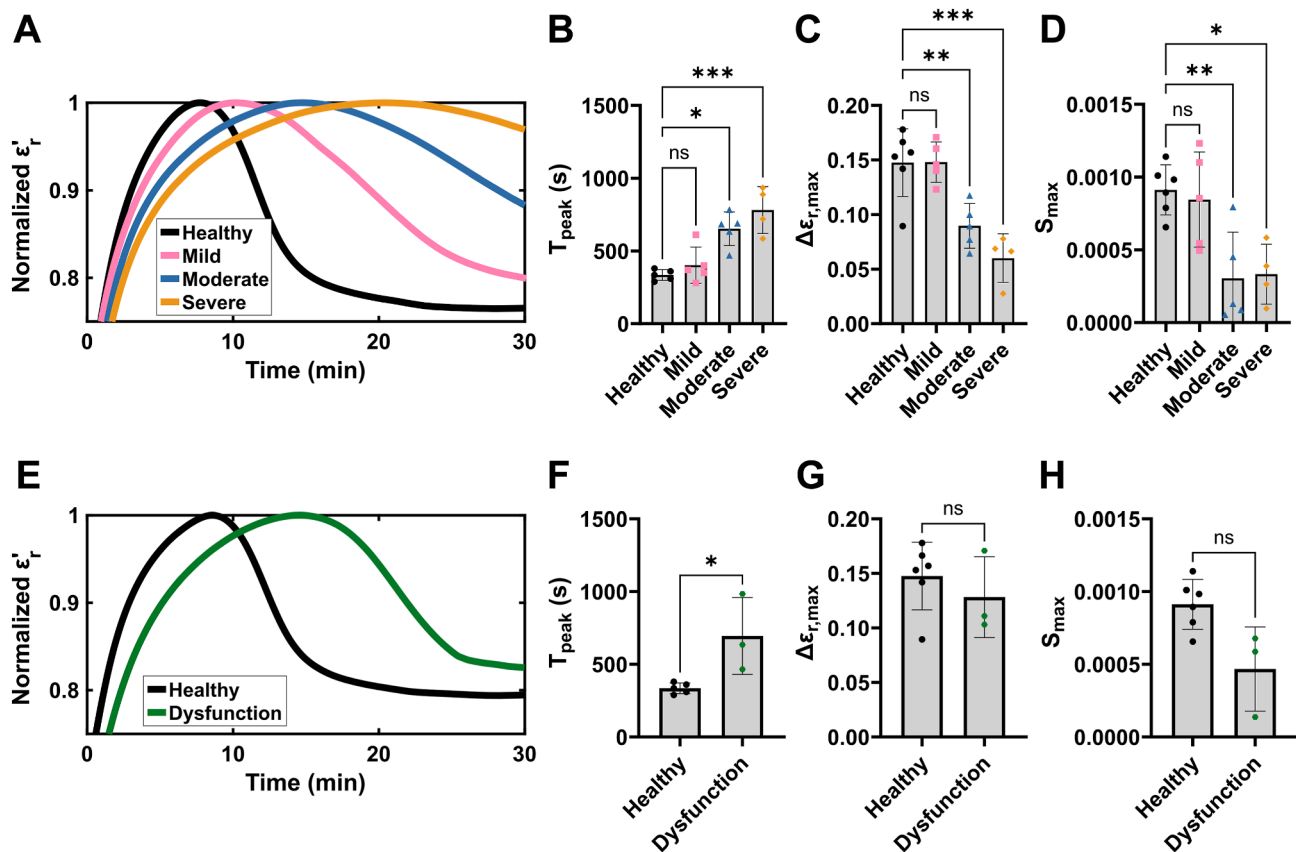


FIGURE 6 Thrombin receptor-activating peptide (TRAP)-6-enhanced dielectric coagulometry was sensitive to platelet defects in pediatric patient blood samples. (A) Representative PlateChek coagulation profiles for a healthy subject and for patients with mild, moderate, or severe thrombocytopenia (TCP; healthy, platelet count > 150 k/ μ L; mild, 70 k/ μ L $<$ platelet count ≤ 150 k/ μ L; moderate, 20 k/ μ L $<$ platelet count ≤ 70 k/ μ L; severe, platelet count ≤ 20 k/ μ L). PlateChek was sensitive to decreased platelet counts in TCP patients. Samples from patients with moderate and severe TCP, but not mild TCP, resulted in significant changes in (B) T_{peak} , (C) $\Delta\epsilon_{r,max}$, and (D) S_{max} . (E) Representative PlateChek coagulation profiles for a healthy subject and a patient presenting with a platelet aggregation dysfunction despite having a healthy platelet count. (F) The primary impact of the platelet aggregation dysfunction was on T_{peak} , which was prolonged 2-fold. There was no significant change in (G) $\Delta\epsilon_{r,max}$ or (H) S_{max} . Statistics are shown as mean $\pm$ SD; ns, not significant; * $P < .05$; ** $P < .01$; *** $P < .001$; **** $P < .0001$.

significantly abnormal in moderate and severe TCP but not in mild TCP (Figure 6B–D). These data corroborate our *in vitro* findings, where we did not observe a statistically significant change in most parameters until platelet counts were ≤ 50 k/ μ L (Figure 3C–E). For patients with platelet aggregation dysfunctions, we found that the primary effect was a statistically significant prolongation in T_{peak} (Figure 6E, F). Although we observed a mild reduction in $\Delta\epsilon_{r,max}$ (Figure 6G) and moderate reduction in S_{max} (Figure 6H), these results were not statistically significant for our limited sample size. Nonetheless, these data provided promising evidence that TRAP-6-enhanced dielectric coagulometry performed by our PlateChek microsensor can predict hemostatic impairment due to both platelet count and function defects in clinically relevant samples.

4 | DISCUSSION

PlateChek performs TRAP-6-enhanced dielectric coagulometry as a whole blood-based approach to directly assess hemostatic

impairment from platelet-specific defects. This differentiates it from other assays that can assess either platelet function or global hemostatic impairment but cannot directly link the 2. For example, platelet aggregometry and PFA systems can be used to assess fibrinogen-mediated platelet aggregation upon agonist stimulation but do not assess blood's ability to perform hemostasis [8]. The T-TAS platelet-specific PL chip quantifies the primary hemostatic potential of platelets upon activation but does not measure secondary hemostasis [33]. While the T-TAS atheroma and hemodilution chips can evaluate secondary hemostasis, they cannot decouple platelet contributions from tissue factor-mediated coagulation [34]. ROTEM and TEG quantify holistic clotting parameters but do not provide diagnostic information specific to platelets [35–37]. The recent TEG Platelet Mapping cartridge has attempted to couple aggregometry with viscoelastometry, but this approach has limited sensitivity and poor reliability [38] and is not predictive of bleeding in patient populations [37]. Compared with these existing assays, which have demonstrated sensitivity to platelet-related hemostatic impairment, the TRAP-6 coating in the PlateChek assay uniquely enables

sensitivity and diagnostic specificity via multiparametric analysis. This can provide significant clinical benefits by broadening the indications for platelet-related intervention or by preventing unnecessary utilization of valuable transfusable platelet products.

There are several other preclinical technologies emerging to address the need for platelet-specific hemostatic-assessment platforms. For example, flow-based microfluidic assays have been developed to measure platelet contractile forces [39], agonist-activated whole blood hemostasis and platelet function under stenosed shear flow [40], or coagulation and platelet function in a wound-mimetic geometry [41]. While flow-based assays can recapitulate hemostasis under physiological shear, they are characterized by elaborate setups, specialized operating procedures, and large blood volume requirements, all of which present obstacles to clinical translation. In contrast, PlateChek utilizes the fully electronic technique of dielectric spectroscopy for hemostatic assessment, which is more readily translatable as it does not rely on complex optical/mechanical parts, pumps to induce flow, or reagent reservoirs. Although adaptable to a flow-based system, if necessary, this technique has demonstrated sensitivity to many aspects of hemostasis even in static or quasistatic conditions. For example, Sekar et al. [42] developed a membrane capacitance sensor with multiparametric sensitivity to platelet count, aspirin, and P2Y₁₂ inhibition, but this assay cannot be performed with whole blood. In another work, Hayashi et al. [43] recently demonstrated the proof-of-concept that dielectric coagulometry can sense antiplatelet therapy in whole blood samples manipulated *in vitro*; however, this study was not validated in patient populations and could not differentiate between different platelet defects [43]. PlateChek obviates the need for blood sample manipulation or reagent addition while improving sensitivity and specificity for platelet-related hemostatic impairment.

Despite the advantages of PlateChek, there remain limitations in the current state of its design. PlateChek is a single-channel system containing a TRAP-6-coated electrode. However, a single-channel system may be insufficient to probe the full spectrum of platelet function. For example, TRAP-6-coated electrodes were insensitive to clopidogrel inhibition, which affects the platelet P2Y₁₂ receptor (Supplementary Figure S14). Because the mechanism of action of TRAP-6-coated electrodes is to induce platelet procoagulant induction via the PAR-1 receptor, a pathway largely independent of P2Y₁₂ activation, this result is not unexpected. However, P2Y₁₂ inhibition is a common antiplatelet therapy approach; this motivates the future development of a multiplexed array of microsensor surface functionalized with judiciously selected bioactive coatings to probe distinct platelet pathways simultaneously. For example, microsensors with electrode surface coated with adenosine diphosphate, arachidonic acid, or convulxin could be developed to probe platelet pathways associated with P2Y₁₂, thromboxane A₂, or GPVI receptors, respectively. Such a parallelized approach could be used to generate a comprehensive assessment of platelet-associated hemostatic potential. Since each microsensor requires < 10 μ L of whole blood volume, a wide array of microsensors could be used while maintaining a smaller blood volume requirement than that of a single

ROTEM, TEG, or aggregometry assay. Moreover, data from all microsensors in the array could be synthesized in multidimensional space to distill the complex dataset into highly specific clinical recommendations. In summary, PlateChek enables sensitive and specific detection of hemostatic impairment due to platelet count or function defects within 15 to 30 minutes using < 10 μ L of whole blood and has potential for development and translation into a powerful diagnostic platform to improve interventional practices in hemostasis and transfusion.

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AUTHOR CONTRIBUTIONS

D.D.: conceptualization, data generation, data analysis, data interpretation, writing, editing, and visualization. S.P.: conceptualization and data generation. C.A.D.: data analysis. S.H., C.A., H.A., and E.S.: data generation. R.J.S.: clinical blood sample collection. M.D.N.: conceptualization, data interpretation, and editing. S.P.A.: clinical blood sample collection, conceptualization, data interpretation, and editing. M.A.S.: conceptualization and data interpretation. P.M. and A.S.G.: conceptualization, data interpretation, writing, editing, and visualization.

DATA AVAILABILITY

The authors declare that all data relevant to this study are available in the main body or supplementary information files.

DECLARATION OF COMPETING INTERESTS

M.A.S. and P.M. are inventors of foundational intellectual property related to this study that has been licensed by Case Western Reserve University to XaTek Inc. They report research funding, consulting fees, and royalties from XaTek Inc. M.D.N. serves as the Chief Medical Officer of Haima Therapeutics and reports personal fees from CSL Behring, Haemonetics, Octapharma, Takeda, and Janssen Pharmaceuticals, as well as grants from Haemonetics, Alexion, and Instrumentation Laboratories. S.P.A. reports research funding from XaTek Inc and PCORI, as well as personal fees from Genentech, CSL Behring, BioMarin, and Novo Nordisk. A.S.G. is a co-founder of Haima Therapeutics and an inventor on patents regarding hemostatic therapeutic technologies that are licensed to Haima Therapeutics. D.D., S.P., C.A.D., S.H., C.A., H.A., E.S., and R.J.S. declare no competing interests.

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SUPPLEMENTARY MATERIAL

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